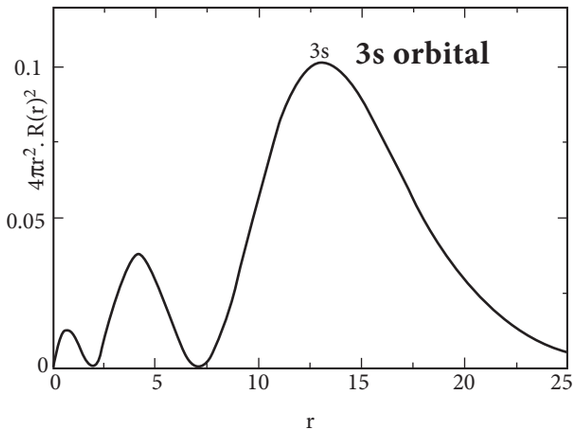


Figure 2.6 (b) - Plot of $4\pi r^2. R(r)^2$ versus r for 3s orbitals of hydrogen

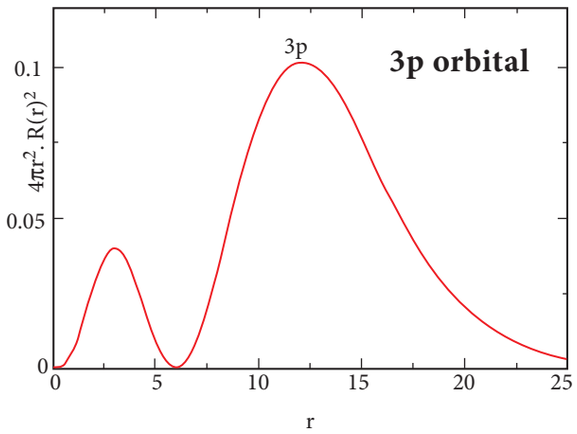


Figure 2.6 (c) - Plot of $4\pi r^2. R(r)^2$ versus r for 3p orbitals of hydrogen

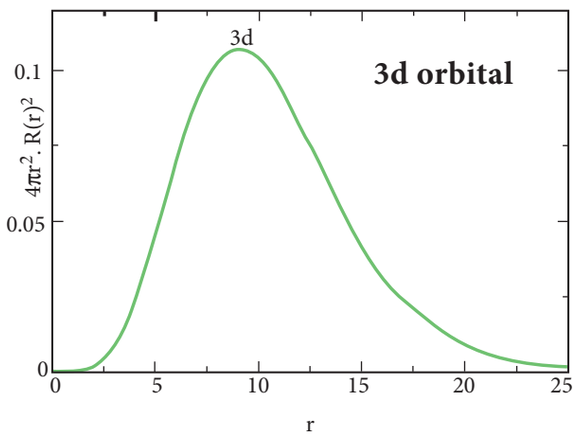


Figure 2.6 (d) - Plot of $4\pi r^2. R(r)^2$ versus r for 3d orbitals of hydrogen

For 2s orbital, as the distance from nucleus r increases, the probability density first increases, reaches a small maximum followed by a sharp decrease to zero and then increases to another maximum, after that decreases to zero. The region where this probability density function reduces to zero is called nodal surface or a radial node. In general, it has been found that ns-orbital has (n–1) nodes. In other words, number of radial nodes for 2s orbital is one, for 3s orbital it is two and so on. The plot of $4\pi r^2. R(r)^2$ versus r for 3p and 3d orbitals shows similar pattern but the number of radial nodes are equal to(n-*l*-1) (where n is principal quantum number and l is azimuthal quantum number of the orbital).

Angular distribution function:

The variation of the probability of locating the electron on a sphere with nucleus at its centre depends on the azimuthal quantum number of the orbital in which the electron is present. For 1s orbital, l=0 and m=0. $f(\theta) = 1/\sqrt{2}$ and $g(\varphi) = 1/\sqrt{2}\pi$. Therefore, the angular distribution function is equal to $1/2\sqrt{\pi}$. i.e. it is independent of the angle θ and φ. Hence, the probability of finding the electron is independent of the direction from the nucleus. The shape of the s orbital is spherical as shown in the figure 2.7

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